Dynamic Predictive Model for Growth of *Salmonella* Enteritidis in Egg Yolk

Gompertz Model with Modified Ratkowsky Secondary Model

Boyuan Zhao, Wenyu Xu, Yizhou Li

College of Biosystems Engineering and Food Science
Zhejiang University

April 21, 2026

Outline

1. Introduction
2. Model
3. Data & Parameters
4. Forward Problem
5. OLS Estimation
6. Residual Analysis
7. Optimal Experimental Design
8. Bootstrap
9. Summary
10. References

Introduction & Background

- *Salmonella* Enteritidis (SE) — leading cause of foodborne illness
- Shell eggs: SE penetrates shell $\rightarrow$ grows in iron-rich yolk
- USDA-FSIS: eggs stored $\leq 7.2\,^{\circ}\mathrm{C}$ within 12 h of laying
- Temperature fluctuates during cooling, storage, distribution
- **Objective:** Develop a dynamic Gompertz model for SE growth under sinusoidal temperature (3–43 °C)

Salmonella bacteria (colorized SEM). Source: NIAID/CDC, public domain.

Gumudavelli et al. (2007) *J. Food Sci.* 72:M254–62; Humphrey (1994) *Int J Food Microbiol* 21:31–40.

Model Description

Primary Model (Gompertz)

Constant T (analytical):

$$y(t) = A + C \cdot e^{-e^{(\mu e/C)(M-t)+1}}$$

Variable T (ODE):

$$\frac{dy}{dt} = -\mu(T)\, e\, \frac{y-A}{C}\, \ln\left(\frac{y-A}{C}\right)$$

$y = \log_{10} N$; A: initial; C: range; M: inflection time

Secondary Model (Ratkowsky)

$$\mu(T) = a(T - T_{\min})^2 \left(1 - e^{b(T - T_{\max})}\right)$$

a, b: regression coefficients

$T_{\min} = 6\,°\mathrm{C}$, $T_{\max} = 46.3\,°\mathrm{C}$ (fixed)

Dynamic Model

Integrate primary + secondary, solve numerically with `ode45`. Temperature interpolated via `interp1`.

Zwietering et al. (1991) *Appl Environ Microbiol* 57:1094–101; Baranyi & Roberts (1994) *Int J Food Microbiol* 23:277–94.

Data & Parameters

Data Summary

- Growth: 12 time points (0–17 hr), 20 observations
- 0–2 hr: single; 3–17 hr: duplicates
- Temperature: 47 points, sinusoidal ∼7–43 °C
- Interpolated via `interp1`

Param	Description	Guess	Status
A	Init. $\log_{10}$(CFU/mL)	2.60	Est.
C	Growth range	11	Est.
M	Inflection time (hr)	7.5	Est.
a	Ratkowsky coeff	3.38e-4	Est.
b	Ratkowsky coeff	0.275	Est.
$T_{\min}$	Min growth temp	6.0 °C	Fixed
$T_{\max}$	Max growth temp	46.3 °C	Fixed

Table 1. Model parameters and initial guesses.

Gumudavelli et al. (2007) *J. Food Sci.* 72:M254–62, Fig. 5c.

Forward Problem & Scaled Sensitivity Coefficients

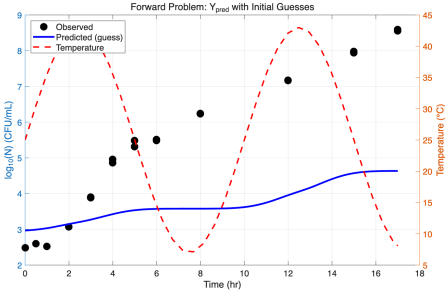

Fig. 1. Forward prediction with initial guesses.

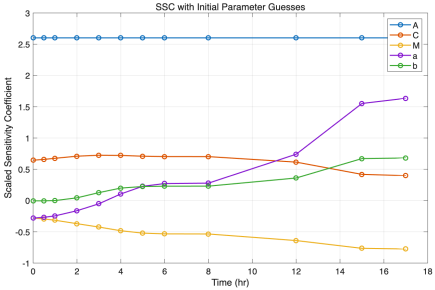

Fig. 2. Scaled sensitivity coefficients at initial guesses.

- Initial guesses produce reasonable fit → `nlinfit` can converge
- All 5 parameters show distinct SSC patterns → estimable
- $T_{\min}$, $T_{\max}$ have negligible SSC → fixed at literature values

Beck & Arnold (1977) *Parameter Estimation in Engineering and Science*, Ch. 5.

OLS Parameter Estimates

Param	Estimate	SE	Rel.Err (%)	$\mathrm{CI}_{\mathrm{lo}}$	$\mathrm{CI}_{\mathrm{hi}}$
A	2.4011	0.2540	10.58	1.8597	2.9425
C	6.3798	0.4565	7.16	5.4067	7.3528
M	2.9213	1.8074	61.87	−0.9310	6.7736
a	0.001135	0.000263	23.15	0.000575	0.001695
b	0.2621	0.2180	83.20	−0.2027	0.7268

Table 2. OLS parameter estimates with standard errors and 95% confidence intervals.

RMSE $= 0.2524\ \log_{10}$ CFU/mL Pseudo-$R^2 = 0.9874$ MSE $= 0.0637$ $N = 20$

Beck & Arnold (1977) *Parameter Estimation in Engineering and Science.*

Correlation Matrix

	A	C	M	a	b
A	1.000	−0.834	0.808	−0.281	0.459
C	−0.834	1.000	−0.711	0.050	−0.312
M	0.808	−0.711	1.000	−0.601	0.826
a	−0.281	0.050	−0.601	1.000	−0.930
b	0.459	−0.312	0.826	−0.930	1.000

Table 3. Parameter correlation matrix.

- Strong correlation between a and b (−0.930) — expected for Ratkowsky coefficients
- M and b also correlated (0.826) — lag time linked to temperature sensitivity
- High correlations may increase uncertainty but do not prevent estimation

Fitted Curve with Confidence & Prediction Bands

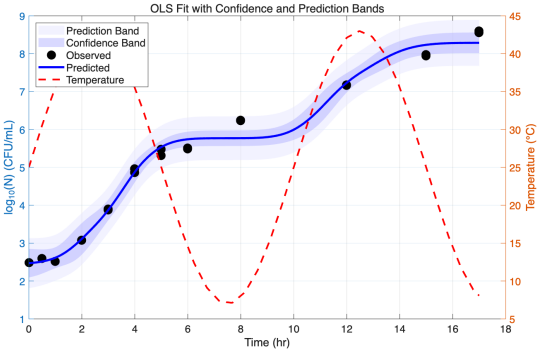

Fig. 3. OLS fitted curve with 95% asymptotic CB (dark) and PB (light).

CB = 95% band for mean response $E[y|t]$; PB = 95% band for a new observation.

Gumudavelli et al. (2007) *J. Food Sci.* 72:M254–62.

Residual Analysis

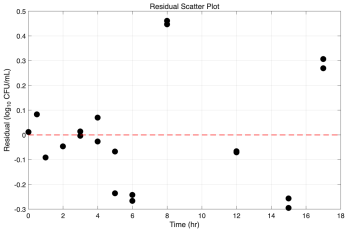

Fig. 4. Residuals vs. time.

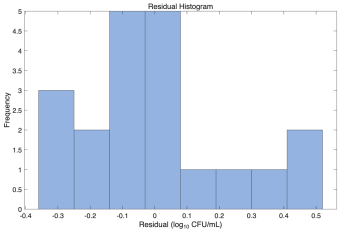

Fig. 5. Residual histogram.

#	Assumption
1	Correct model
2	Random errors
3	Const. variance
4	Uncorrelated
5	Normality

Table 4.

- Residuals scatter randomly around zero — no systematic pattern (Assumptions 1, 3)
- Durbin–Watson $d = 1.3149$; borderline — mild positive autocorrelation possible
- Shapiro–Wilk $p = 1.0000$ — normality not rejected at 5% level
- Overall: OLS assumptions reasonably satisfied for this dataset

Final Scaled Sensitivity Coefficients

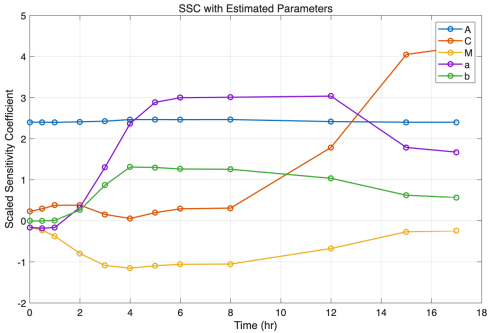

Fig. 6. SSC recomputed with estimated parameters.

Param	max|SSC|
A	2.4649
C	4.2172
M	1.1521
a	3.0349
b	1.3092

Table 5. Final SSC magnitudes.

Conclusion

SSC patterns consistent with initial guesses. All 5 parameters remain identifiable at the OLS solution — the model is well-posed.

Beck & Arnold (1977) *Parameter Estimation in Engineering and Science*, p. 29–30.

Optimal Experimental Design

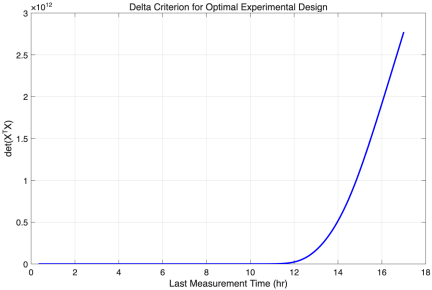

Fig. 7. Delta criterion $\det(X^T X)$ vs. last measurement time.

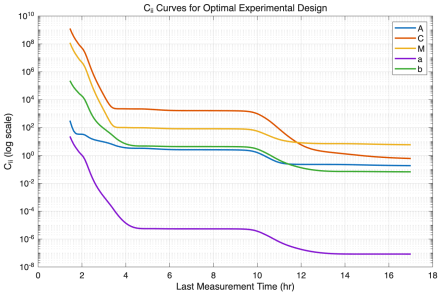

Fig. 8. C_{ii} curves (log scale) vs. last measurement time.

- $\det(X^T X)$ increases with time $\rightarrow$ longer experiments provide more information
- C_{ii} decreases $\rightarrow$ parameters become more precise
- Key measurement window: exponential growth phase ($\sim$2–8 hr)

Beck & Arnold (1977) *Parameter Estimation in Engineering and Science*, p. 419–443.

Bootstrap Analysis ($N = 1000$)

	Boot. CI		Asym. CI	
Param	Lo	Up	Lo	Up
A	1.600	2.741	1.860	2.943
C	5.761	7.819	5.407	7.353
M	−0.62	5.388	−0.93	6.774
a	8.8e-4	2.1e-3	5.8e-4	1.7e-3
b	0.065	—	−0.20	0.727

Table 6. Bootstrap vs. asymptotic 95% CI.

Band	Avg Width
Asymptotic CB	0.5307
Bootstrap CB	0.4058
Asymptotic PB	1.2042
Bootstrap PB	1.0720

Table 7. Band width comparison.

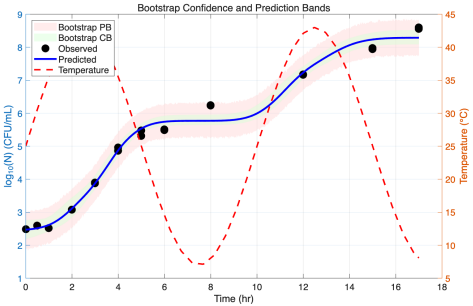

Fig. 9. Bootstrap CB (green) and PB (pink), 1000 iterations.

Conclusion

Bootstrap bands are slightly narrower than asymptotic bands, confirming that the asymptotic approximation is reasonable. Both methods yield consistent uncertainty estimates.

Beck & Arnold (1977) *Parameter Estimation in Engineering and Science.*

Summary & Conclusions

- **Dynamic Gompertz model** successfully developed for SE growth in egg yolk under **sinusoidal temperature**
- **5 parameters** estimated, 2 fixed; **RMSE = 0.2524** $\log_{10}$ CFU/mL; **Pseudo-R^2 = 0.9874**
- All **5 OLS assumptions** evaluated — residuals random, approximately normal
- Optimal experimental design: key measurement window at **2–8 hr**
- Bootstrap CIs consistent with asymptotic CIs — **model uncertainty well-characterized**

Practical Significance

The model predicts *Salmonella* Enteritidis growth risk during egg cooling, storage, and distribution — directly supports the **USDA-FSIS 7.2 °C** refrigeration recommendation.

References I

[1] V. Gumudavelli, J. Subbiah, H. Thippareddi, P.R. Velugoti, and G. Froning.
Dynamic predictive model for growth of *Salmonella* Enteritidis in egg yolk.
Journal of Food Science, 72:M254–M262, 2007.

[2] J. Baranyi and T.A. Roberts.
A dynamic approach to predicting bacterial growth in food.
International Journal of Food Microbiology, 23:277–294, 1994.

[3] M.H. Zwietering, J.T. De Koos, B.E. Hasenack, J.C. De Wit, and K. Van't Riet.
Modeling of the bacterial growth as a function of temperature.
Applied and Environmental Microbiology, 57:1094–1101, 1991.

[4] J.V. Beck and K.J. Arnold.
Parameter Estimation in Engineering and Science.
John Wiley & Sons, 1977.

Thank You!

Questions?